Certification Project

Problem Statement

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AppleBite Co. is using the cloud for one of its products. The project uses modular components and multiple frameworks, and the company wants the components to be developed by different teams or third-party vendors.

The company's goal is to deliver product updates frequently to production with high quality and reliability. They also aim to accelerate software delivery speed, enhance quality, and reduce feedback time between developers and testers.

As development progressed, they faced multiple problems due to the various technologies involved in the project. The following are the problems:

- Building complex builds is difficult
- Incremental builds are difficult to manage and deploy

To solve these problems, they need to implement Continuous Integration and Continuous Deployment with DevOps using the following tools:

Git – For version control for tracking changes in the code files

Jenkins – For continuous integration and continuous deployment

Docker – For deploying containerized applications

Ansible - Configuration management tools

This project focuses on deploying code to dev, stage, or prod environments with just a single button click.

Link for the sample PHP application: https://github.com/edureka-devops/projCert.git

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Business Challenge/Requirement

As soon as the developer pushes the updated code on the GIT master branch, a new test server should be provisioned with all the required software. Post this, the code should be containerized and deployed on the test server.

The deployment should then be built and pushed to the prod server.

All this should happen automatically and should be triggered from a push to the GitHub master branch.

Steps for Executing the Solution:

- Use Master VM for Jenkins, Ansible, GIT, etc.
- Use the fresh instance for Jenkins Slave Node (Test Server)
- Change the IP address of the VMs accordingly
- Add Build Pipeline Plugin and Post-build task plugin to Jenkins on the master VM
- Install Python, OpenSSH server, and Git on the slave node manually.
- Use the image devopsedu/webapp and add your PHP website to it with a Dockerfile.
- Push the PHP website and the Dockerfile to a Git repository

The below tasks should be automated through Jenkins by creating a pipeline:

1. Install and configure the puppet agent on the slave node (Job 1)
2. Push an Ansible configuration on the test server to install docker (Job 2)
3. Pull the PHP website, and the Dockerfile from the git repo and build and deploy your PHP docker container. (Job 3)
4. If Job 3 fails, delete the running container on the Test Server

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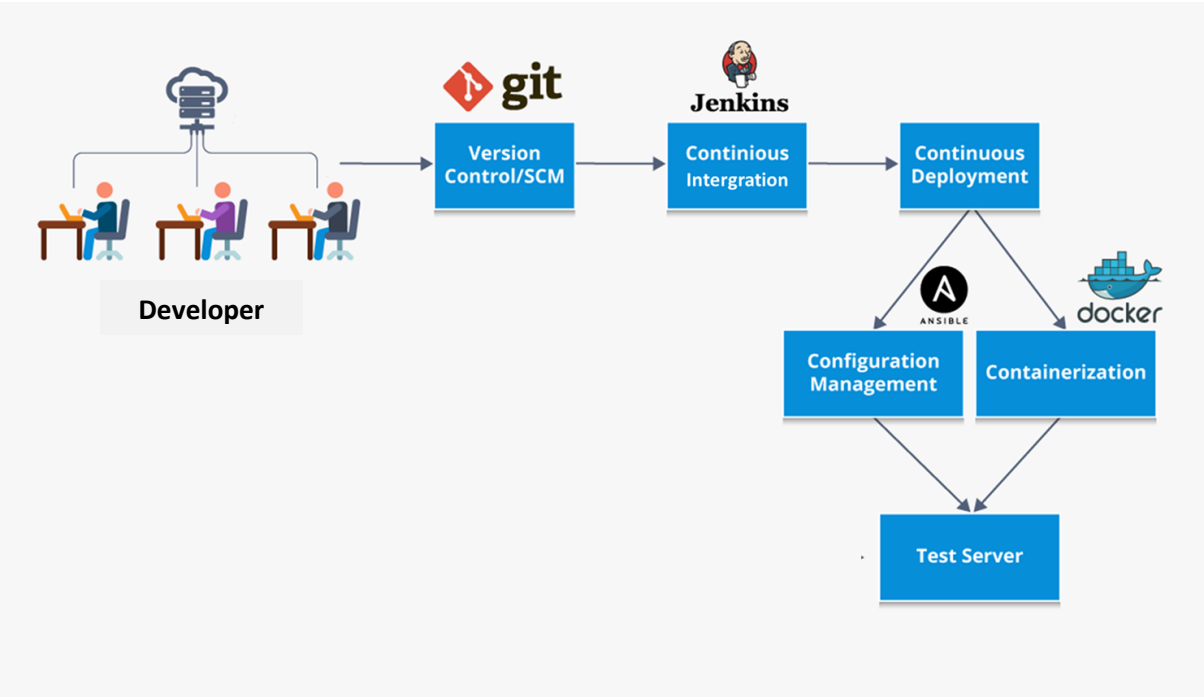

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